

# Dishank Oza

+1 (669) 281-9480 | Santa Clara, CA | dishankwork@gmail.com | linkedin.com/in/dishankoz

## EDUCATION

Masters in Computer Science - Santa Clara University | 4.0/4.0

Sep 2024 - Mar 2026

Bachelors in Computer Engineering - University of Mumbai | 3.5/4.0

Aug 2017 - Jun 2021

Relevant Coursework: Distributed Computing, Machine Learning, Advance OS, Database Systems, Data Structures, Advanced Algorithms.

## PROFESSIONAL EXPERIENCE

### Teradata

San Diego, USA

Software Engineer Intern II

Jun 2025 - Sept 2025

- Upgraded Teradata's database engine from **LLVM 16** → **20** by removing typed pointers, deprecated APIs, and redundant instruction casts, while updating codegen/JIT interfaces for compatibility and performance.
- Built a custom IR-analysis and diffing tool to compare LLVM IR across versions, measure key attributes (GEP depth, nuw, initializes(...), vectorization hints, inlining, aliasing), and visualize the results through charts—revealing structural optimizations that contributed to an observed **~10 % CPU-time** reduction on 1 TB workloads (Teradata VM and AWS).

### Freespace

Mumbai, India

Senior Software Engineer

May 2023 - Aug 2024

- Led a team of four** in Agile development to deliver business-critical features and implement CI/CD pipelines with Jenkins, earning recognition as **Freespace Captain** for exceptional leadership and problem-solving.
- Architected and developed a **microservice** to streamline booking events using **AWS DMS**, **Kinesis**, and **EventBridge** to distribute event generation from **30 to 1 events per user daily**, optimizing resource usage and system efficiency.
- Revamped API transaction handling by replacing serialized locks with **Redis distributed locks** in a Spring Boot application, reducing database load and improving concurrency, cutting application initial load time from **30 to 9s** during peak times.
- Pioneered scalable system architecture by designing and implementing the migration of booking APIs to an **event-driven** architecture with AWS SQS FIFO and Lambdas, **reducing server count from 12 to 8** during peak time.

### Freespace

Mumbai, India

Software Engineer

Jun 2021 - May 2023

- Led migration to single page application (**SPA**) with **React routing**, route-based and custom **Webpack code splitting**, **HOCs**, doubling first load speed and improving repeat visits by **100%**.
- Developed a custom frontend component library created with **React**, **TypeScript**, and Storybook, delivering tailored branding styles for clients and boasting UI consistency and development efficiency.
- Designed and implemented a serverless notification system leveraging **AWS Lambda**, **SQS**, **SNS**, and SES for scalability, with multi-channel support (email, push, SMS) and a **CloudFormation** stack for automated infrastructure provisioning.
- Developed RESTful APIs using Spring Boot and GraphQL for an enterprise application serving 100,000+ users across 30+ global clients, managing up to **5,000** concurrent requests per minute.

## PROJECTS

LogSight [github] | Java, Spring Boot

- Designed and implemented a distributed log aggregation platform using Spring Boot microservices, Kafka, and the ELK stack (Elasticsearch, Logstash, Kibana), featuring trace correlation, Docker-based deployment, and end-to-end observability through Prometheus, Grafana, and CI/CD automation.

AmberDB – Distributed SQL Database [github] | Go, SQL

- Designed and built a scalable, fault-tolerant SQL database with Raft-based replication, MVCC using Hybrid Logical Clocks, dynamic sharding, and 2PC for atomic cross-range transactions.

Spacelink [github] | Next.js, Node

- Developed a 2D virtual space platform for real-time avatar interactions, with WebSockets and Protobuf for efficient data transfer. Integrated video calls via WebRTC and managed data with PostgreSQL, using TypeScript, Prisma, and Next.js.

## TECHNICAL SKILLS

**Programming & Database:** Java, JavaScript, Typescript, Go, C/C++, HTML/CSS, PostgreSQL/MySQL, Cassandra, MongoDB, Neo4j, Redis

**Libraries/Framework/Tools:** React.js, Next.js, Vue Node.js, Tailwind, Spring Boot, Git, Jenkins, Jest, JIRA, Postman, Protocol Buffers, GraphQL

**Cloud Technologies & Others:** AWS, Firebase, Terraform, Docker, Kubernetes/EKS, Ansible, Grafana, Prometheus, Linux, Hadoop

**Development Practices:** Full-stack development, CI/CD, Automation, TDD, Design Patterns, Agile, SDLC, SOLID, and Code Reviews

**Certifications & Publications:** AWS Certified Developer Associate [Link], Scrum Alliance-Scrum Foundations, Discovering Backend Bottlenecks, Gesture-Based Interaction System [Link]

## ACHIEVEMENTS, AWARDS, AND LEADERSHIP

- Awarded Freespace Captain for successfully leading the team, utilizing exceptional problem-solving skills, and ensuring timely delivery of project deliverables
- 3rd in Smart India Hackathon qualifier 2020 for license plate recognition and automated parking system using Raspberry Pi.
- Member of Team Vaayushastra, ranked 9th globally and 2nd in the Asia-Pacific region in the SAE Aero Design Competition 2020.